

Ivan Nikolaev

Product Engineer

Senior backend in Go and Ruby on Rails, AI-native development. Almost 7 years shipping.
Moscow, remote.

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1.57 → 0.49 review comments per MR 40 → 104 merged MRs per half-year
2 → 6 tasks per sprint, median

Main work

Context for agents, not prompts

AI in engineering, day job, 2026

Wired the tracker, wiki and GitLab into the agent via my own MCP servers, wrote ~15 skills for daily chores, set up context hygiene and measured everything.

1.57 → 0.49 review comments per MR; 71% → 83% MRs with zero comments

A RAG bot over the wiki, in the team messenger

Day job, in production, 2026

A Go bot over 1,834 wiki pages and OpenAPI specs. Added golden-set evals, found a silent table bug corrupting 57% of pages, fixed silent degradation of long threads.

13 / 15 → 16 / 17 golden-set eval; 7K, answers silently degraded → 24K plus a recap on overflow context in long threads, tokens

“Review 540”: performance reviews out of Excel

Side project, B2B SaaS, 16 days

Moved a 540° review methodology from Excel into a Go and React B2B service in 16 days: HMAC pseudonymization, multi-tenancy, 94 MCP tools, mutation testing.

hand-made Excel formulas → Excel reference locked in a regression test scoring; person id next to the score → HMAC, unrecoverable after retention answer-to-respondent link

“Svod”: documents where styling lives in the template

Side project, closed beta, svod.webtm.ru

A document editor where styling lives in the template. One AST and one renderer for PDF, DOCX, HTML and Markdown. 43 MCP tools: an agent turns a Word contract into a template.

in every paragraph → in one template formatting; 214 → fixed in waves audit findings (19 dimensions)

Open source: provenance and database-seed-cli

My own libraries, Ruby and Go, 2026

provenance, an audit gem for Rails, 450 downloads. database-seed-cli in Go, a consistent data generator for PostgreSQL, tested on databases with up to 417 tables.

rewritten in every service → gem "provenance" auditing in a Rails app; prod dump or by hand → ~82K rows, zero manual edits dev data

Other work

- **EDUG, a learning bot.** an idea → a working bot and 2nd place
- **Hermes, a home agent.** 10+ home service web UIs → one Telegram chat, 20 MCP servers
- **CI tests 2.6× faster.** ~12.5 min → ~4.5 min tests in the pipeline
- **Flaky tests show up on the MR.** re-run the pipeline and forget → 15 flaky specs caught on 75 MRs
- **An MCP server for the tracker and wiki.** agent can't see tasks → 22 tools
- **Agent-driven defect analysis.** by hand, quarterly → ~\$0.17 per 24 tasks
- **Read replicas without surprises.** 805 → 0 violations over 17,207 tests
- **Document-first API tests.** schema from tests → tests from schema
- **API descriptions across 10 services.** 0 → 1 100+ parameter descriptions
- **Redis over TLS in 10 services.** token blacklist silently broken → 10 services on TLS, blacklist works
- **Splitting a shared bucket.** 1 bucket, 78 GB → one bucket per service
- **Real errors visible in Sentry again.** errors buried under zombie jobs → you see what needs fixing
- **Delivery report.** manual Excel → Grafana

Experience

Nov 2023 – now

Senior Backend Engineer, Lead Software Engineer, ПСБ цифровая лаборатория

An education platform, ~30 services in Ruby on Rails and Go, Kafka, PostgreSQL with replicas, k8s. 427 resolved tasks, 346 MRs, third by commits of 59 authors in the main service. Code review, buddy for new hires.

Aug 2020 – Nov 2023	Full-stack Developer, ТренингСпэйс Client projects; a year in, mentor on the largest one: estimates and requirements together with the PM. Later a product team and backend mentoring for the mobile team.
Feb 2022 – Jun 2023	Lecturer, alongside the main job, гбпоу тпск им в.м максимчука Databases, information systems, operating systems. Supervised graduation projects of 20 students, coached students for championships and hackathons.
Oct 2022 – Dec 2022	Instructor, Альтшу Университет Git, programming basics, database design.
Dec 2019 – Aug 2020	Intern developer → Technician, ФГАНУ ЦИТиС Intern developer, then technician. My first Ruby on Rails service, a support desk with server-side rendering.

Skills

Core	Go, Ruby on Rails, PostgreSQL, Kafka, Redis, Sidekiq, REST, GraphQL, OpenAPI
Infra	Docker, Kubernetes, GitLab CI, MinIO, OpenSearch, Grafana, Sentry
Frontend when needed	TypeScript, React
Team	code review, buddy for new hires, backend mentoring, supervised graduation projects of 20 students, WorldSkills coaching
Languages	Russian native, English B1: docs and written communication
AI at work	LLM, AI agents, Claude Code, MCP, tool calling, RAG, evals, custom skills

Championships

WorldSkills Russia, “IT Software Solutions for Business”, 2017–2019. 1 place, Moscow championship; 5 place, national final; 4 place, DigitalSkills. As an expert: compatriot expert at WorldSkills: coached and accompanied 4 competitors; twice invited industry expert at Moscow championships; 50+ demonstration exams as expert and chief expert.

Games

Papenture (Unreal Engine, Unreal MegaJam 2022, a team of 6); **AV** (GameMaker, Major Jam 3, theme “Retro”); **Bull Hellsing** (GameMaker, in development); **Tagman** (Action platformer, in development). itch.io/zander-games

Education

2021–2025	Национальный исследовательский университет «МЭИ» Institute of Radio Engineering and Electronics
2016–2021	ESEDU Computer Science and Engineering
2019	Courses C# 6.0, Microsoft M10262A

Full version with case studies and screenshots: nikolaevyiu.ru. Habr, August 2026, 6.3K reads: habr.com/ru/articles/1075224/